

ITA0612-Machine Learning

CO3 - AT1 EXPERIMENTS

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Experiment 1: Naïve Bayes Classifier on Iris Dataset

Import Required Libraries

Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
```

Load the Iris Dataset

Code

```
iris = load_iris()

X = iris.data
y = iris.target
```

Split the Dataset into Training and Testing Sets

Code

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

Train the Gaussian Naïve Bayes Model

```
model = GaussianNB()
```

```
model.fit(X_train, y_train)
```

▼ GaussianNB ⓘ ?

```
GaussianNB()
```

Predict the Test Data

```
y_pred = model.predict(X_test)
```

```
print(y_pred)
```

```
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 2 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]
```

Calculate Accuracy

```
from sklearn.metrics import accuracy_score
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("Accuracy:", accuracy)
```

```
Accuracy: 0.9777777777777777
```

Calculate Precision

Code

```
from sklearn.metrics import precision_score
```

```
precision = precision_score(y_test, y_pred, average='weighted')  
print("Precision:", precision)
```

Precision: 0.9793650793650793

Calculate Recall

```
from sklearn.metrics import recall_score
```

```
recall = recall_score(y_test, y_pred, average='weighted')  
print("Recall:", recall)
```

Recall: 0.9777777777777777

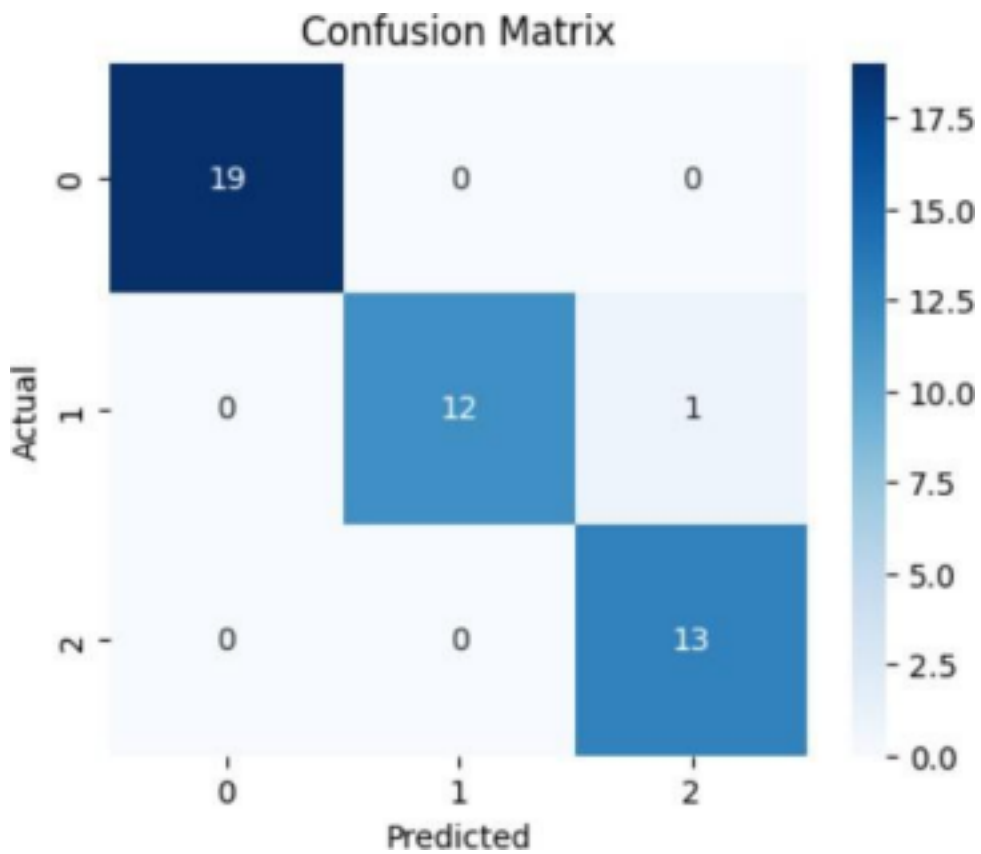
Classification Report

```
from sklearn.metrics import classification_report
```

```
print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	19
1	1.00	0.92	0.96	13
2	0.93	1.00	0.96	13
accuracy			0.98	45
macro avg	0.98	0.97	0.97	45
weighted avg	0.98	0.98	0.98	45

Confusion Matrix (All Features)



Experiment Using Sepal Features

Train Model Using Sepal Features

```

#only sepal features
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, precision_score, recall_score

iris = load_iris()

X = iris.data[:, :2]      # Sepal Length and Sepal Width
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = GaussianNB()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Sepal Features")
print("Accuracy :", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred, average='weighted'))
print("Recall   :", recall_score(y_test, y_pred, average='weighted'))

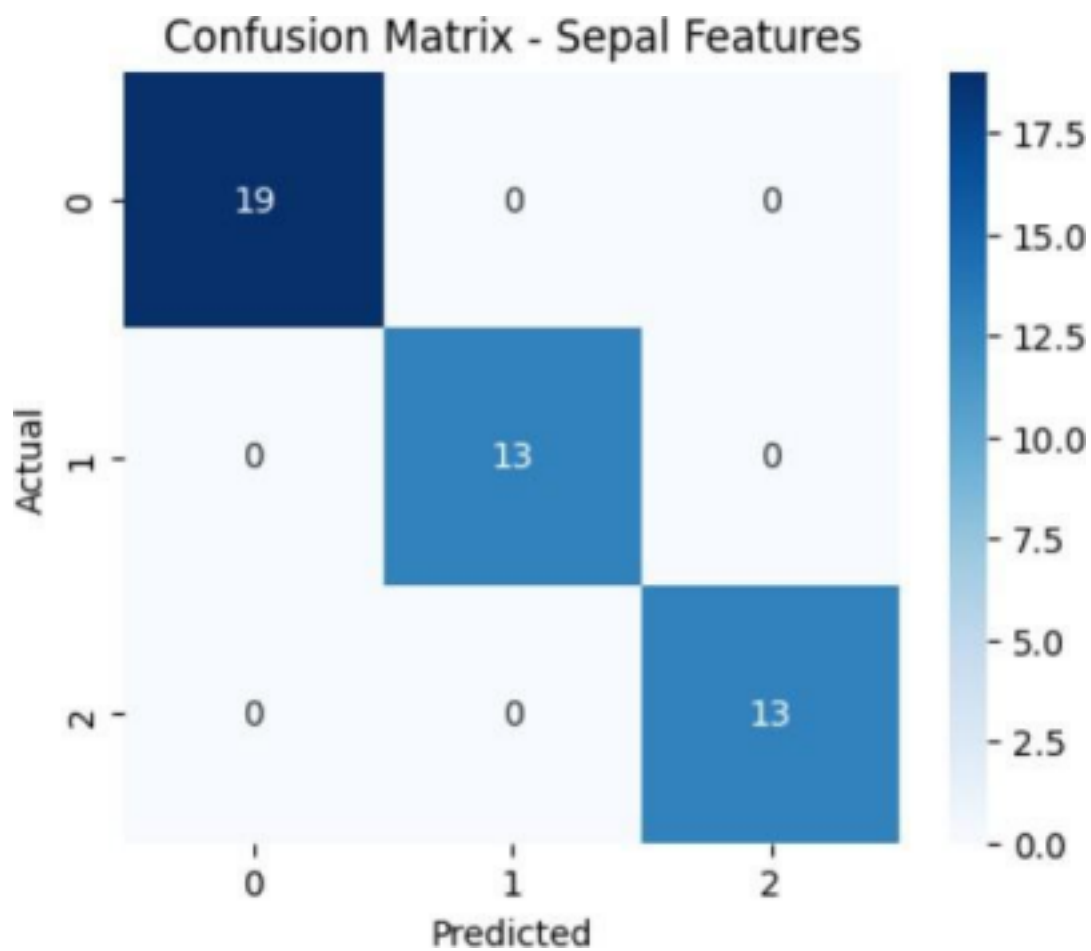
```

```

Sepal Features
Accuracy : 0.8222222222222222
Precision: 0.833841684822077
Recall   : 0.8222222222222222

```

Confusion Matrix (Sepal Features)



Experiment Using Petal Features

Train Model Using Petal Features

```

#only petal features
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, precision_score, recall_score

iris = load_iris()

X = iris.data[:, 2:]      # Petal Length and Petal Width
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = GaussianNB()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Petal Features")
print("Accuracy :", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred, average='weighted'))
print("Recall   :", recall_score(y_test, y_pred, average='weighted'))

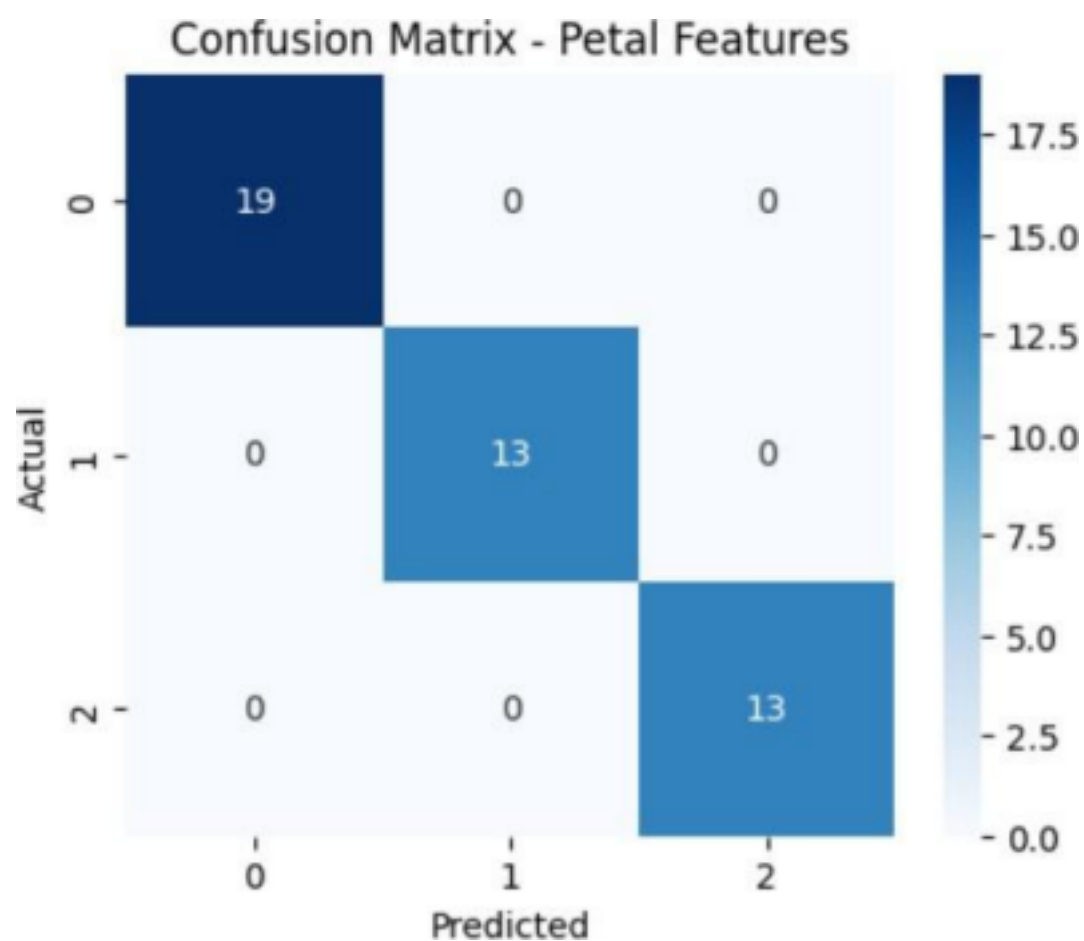
```

```

Petal Features
Accuracy : 1.0
Precision: 1.0
Recall   : 1.0

```

Confusion Matrix (Petal Features)



Experiment Using Smoothing

Compare Different Smoothing Values



Performance Comparison Table

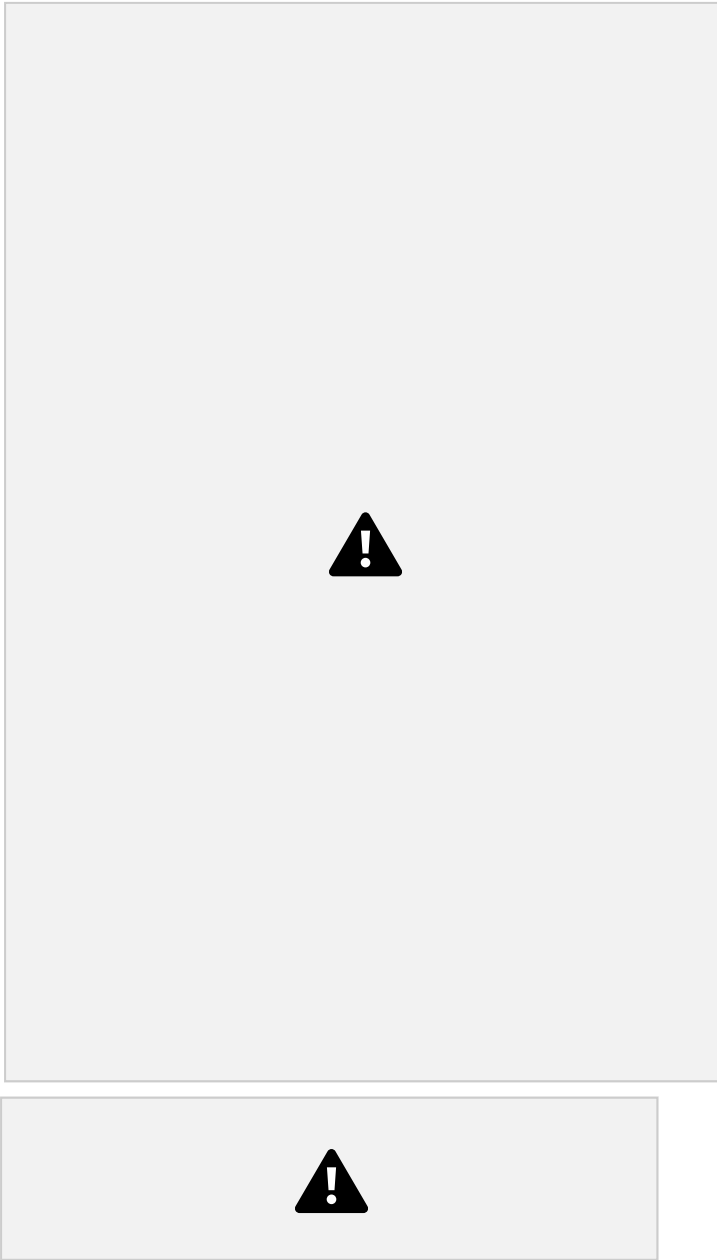


Experiment 2: MLE vs Bayesian

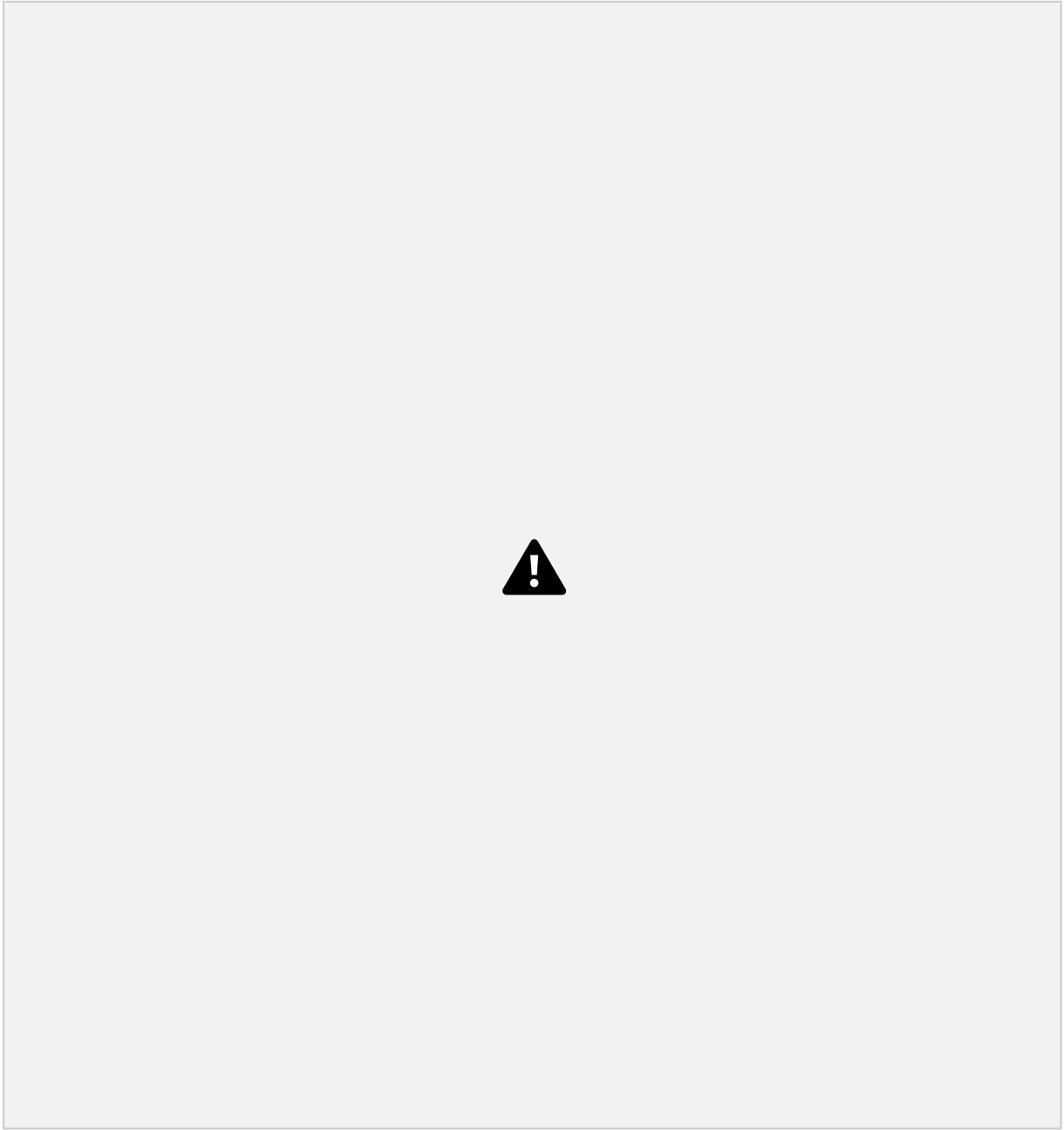
Estimation Load Iris Dataset



Small Sample (20 Samples)



Large Sample (150 Samples)



Histogram



Comparison Table

